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Cooling of photovoltaic cells under concentrated illumination: a critical review

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Abstract

Cooling of photovoltaic cells is one of the main concerns when designing concentrating photovoltaic systems. Cells may experience both short-term (efficiency loss) and long-term (irreversible damage) degradation due to excess temperatures. Design considerations for cooling systems include low and uniform cell temperatures, system reliability, sufficient capacity for dealing with 'worst case scenarios', and minimal power consumption by the system. This review presents an overview of various methods that can be employed for cooling of photovoltaic cells. It includes the application to photovoltaic cells of cooling alternatives found in other fields, namely nuclear reactors, gas turbines and the electronics industry. Different solar concentrators systems are examined, grouped according to geometry. The optimum cooling solutions differ between single-cell arrangements, linear concentrators and densely packed photovoltaic cells. Single cells typically only need passive cooling, even for very high solar concentrations. For densely packed cells under high concentrations (>150 suns), an active cooling system is necessary, with a thermal resistance of less than $10^{-4}\,\mathrm{K\,m^2/W}$. Only impinging jets and microchannels have been reported to achieve such low values. Two-phase forced convection would also be a viable alternative.
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Keywords: Solar concentration; Photovoltaics; Cooling; Literature review

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1. Introduction

1.1. Cooling requirements for concentrator cells

Concentration of sunlight onto photovoltaic cells, and the consequent replacement of expensive photovoltaic area with less expensive concentrating mirrors or lenses, is seen as one method to lower the cost of solar electricity. Because of the reduction in solar absorber area, more costly, but higher efficiency PV cells may be used. However, only a fraction of the incoming sunlight striking the cell is converted into electrical energy (a typical efficiency value for concentrator cells is 25% [1]). The remainder of the absorbed energy will be converted into thermal energy in the cell and may cause the junction temperature to rise unless the heat is efficiently dissipated to the environment. The major design considerations for cooling of photovoltaic cells are listed below:

Cell temperature. The photovoltaic cell efficiency decreases with increasing temperature [2–4]. The cells will also exhibit long-term degradation if the temperature exceeds a certain limit [5,6]. The cell manufacturer will generally specify a given temperature degradation coefficient and a maximum operating temperature for the cell.

Uniformity of temperature. The cell efficiency is known to decrease due to non-uniform temperatures across the cell [7–11]. In a photovoltaic module, a number of cells are electrically connected in series, and several of these series connections can be connected in parallel. Series connections increase the output voltage and decrease the current at a given power output, thereby reducing the ohmic losses. However, when cells are connected in series, the cell that gives the smallest output will limit the current. This is known as the 'current matching problem'. Because the cell efficiency decreases with increasing temperature, the cell at the highest temperature will limit the efficiency of the whole string. This problem can be avoided through the use of bypass diodes [12] (which bypass cells when they reach a certain temperature—in this arrangement you lose the output from this cell, but the output from other cells is not limited) or by keeping a uniform temperature across each series connection.

Reliability and simplicity. To keep operational costs to a minimum, a simple and low maintenance solution should be sought. This also includes minimising the use of toxic materials due to health and environmental concerns. Reliability is another important aspect because a failure of the cooling system could lead to the destruction of the PV cells. The cooling system should be designed to deal with 'worst case scenarios' such as power outages, tracking anomalies and electrical faults within modules [6].

Useability of thermal energy. Use of the extracted thermal energy from cooling can lead to a significant increase in the total conversion efficiency of the receiver [13]. For this reason, subject to the constraints above, it is desirable to have a cooling system that delivers water at as high a temperature as possible. Further, to avoid heat loss through a secondary heat exchanger, an open-loop cooling circuit is an advantage.

Pumping power. Since the power required of any active component of the cooling circuit is a parasitic loss [13], it should be kept to a minimum.

Material efficiency. Materials use should be kept down for the sake of cost, weight and embodied energy considerations.

1.2. Concentrator geometries

It is sensible to distinguish between concentrators according to their method for concentrating (mirrors or lenses), concentration level or geometry. In this review, concentrators will be grouped according to geometry, because the requirements for cell cooling differ considerably between the various types of concentrator geometries. The issue of shading, however, is different for lens and mirror concentrators. If lenses are used, the cells are normally placed underneath the light source, and so shading by the cooling system does not occur. For mirror systems, the cells are generally illuminated from below, which makes shading an important issue to consider when designing the cooling system. Concentrators can be roughly grouped as in the following sub-sections.

1.2.1. Single cells

In small point-focus concentrators, sunlight is usually focused onto each cell individually. This means that each cell has an area roughly equal to that of the concentrator available for heat sinking, as shown in Fig. 1. A cell under $50\times$ concentration should have 50 times its area available for spreading of heat. This geometry means passive cooling can be used at quite high concentration levels (see Section 3.1). Single cell systems commonly use various types of lenses for concentration. Another variant is where reflective concentrators transmit the concentrated light through optical fibers onto single cells.

1.2.2. Linear geometry

Line focus systems typically use parabolic troughs or linear Fresnel lenses to focus the light onto a row of cells. In this configuration, the cells have less area available for heat sinking because two of the cell sides are in close contact with the

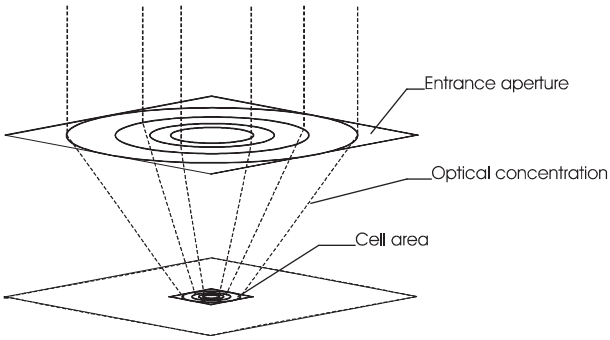


Fig. 1. Single-cell concentrator: dashed line shows area available for heat sinking.

neighbouring cells, as shown in Fig. 2. The areas available for heat sinking extend from two of the sides and the back of the cell.

1.2.3. Densely packed modules

In larger point-focus systems, such as dishes or heliostat fields, the receiver generally consists of a multitude of densely packed cells. The receiver is usually placed slightly away from the focal plane to increase the uniformity of illumination. Secondary concentrators (kaleidoscopes) may be used to further improve flux homogeneity [14]. Densely packed modules present greater problems for cooling than the two previous configurations discussed, because, except for the edge cells, each of the cells only has its rear side available for heat sinking, as shown in Fig. 3. This means that, in principle, the entire heat load must be dissipated in a direction normal to the module surface. This generally implies that passive cooling cannot be used in these configurations at their typical concentration levels.

1.3. Heat transfer coefficients and thermal resistances

The commonly used quantities for comparing the heat transfer characteristics of cooling systems are heat transfer coefficients h or thermal resistances R. These can be defined in several different ways depending on the application. When dealing with passive cooling systems, h is generally defined as

$$h = \frac{\dot{q}}{T_s - T_0}, \tag{1}$$

where $\dot{q}$ is the heat input per unit area, T_s is the mean surface temperature, and T_0 is the ambient temperature. R, when used per unit area, is just the inverse of h. In the

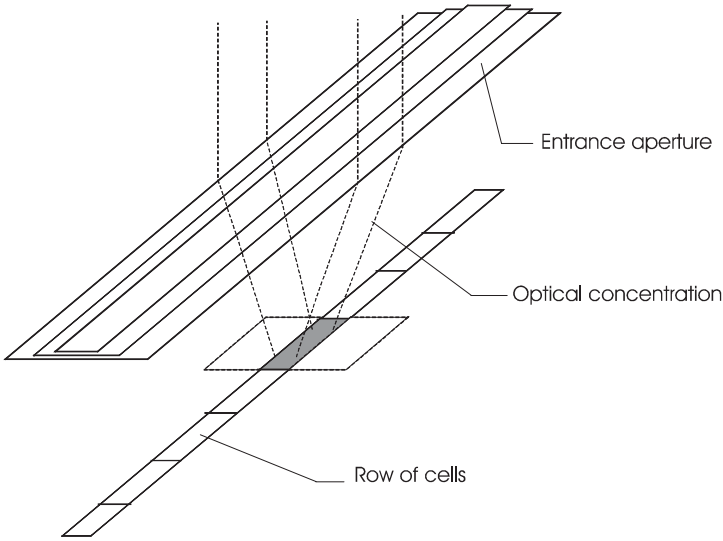


Fig. 2. Linear concentrator: dashed lines show area available for heat sinking.

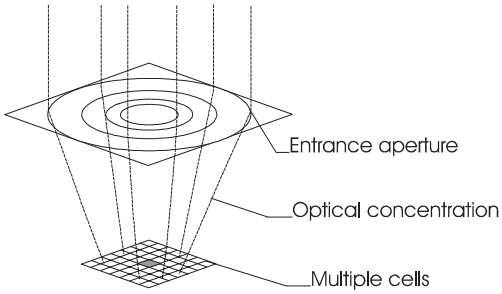


Fig. 3. Densely packed cells: area available for cooling is only the rear side of the cell.

case of single-phase forced convection cooling, one will generally use a local heat transfer coefficient

$$h = \frac{\dot{q}}{T_w - T_f}, \tag{2}$$

where T_w and T_f are the mean wall and fluid temperatures at any given point. For natural convection, boiling and radiative heat transfer, $\dot{q}$ is not proportional to ΔT, and therefore R and h vary with temperature [15]. In the case of radiation, a simplification is often used to linearise the calculation (given in Section 2.1). The literature sometimes quotes values for h or R with natural convection or two-phase forced convection, and these are included in this article. However, these should be interpreted with caution and not be assumed to be valid for a large range of temperatures.

2. One-dimensional thermal model of cell and encapsulation layers

To examine the best cooling system for a given concentrator requires the development of a thermal model that will predict the heating and electrical output of cells. In this review, a one-dimensional model is used because this is consistent with a closely packed set of cells where heat flow is primarily directed in the normal direction. Models for other layouts can be easily extended from this model, or they can be found in literature, e.g. Ref. [2]. Models for single-cell point focus are described in Refs. [9,16,17] and for linear geometry in Refs. [7,11,18]. The idealised cell and its mounting is shown schematically in Fig. 4, where I is the incoming concentrated solar flux, and t_g, t_a, t_c, t_{so} and t_s denote the thicknesses of the various layers.

This configuration can be represented by the equivalent thermal circuit shown in Fig. 5, where R denotes a thermal resistance. Note that because this model is one-dimensional, all relevant values are per unit area: the units of R are [K m^2/W] while the units of $\dot{q}$ are [W/m^2]. T_g, T_s and T_0 are the temperatures of the top surface of the cover glass, the bottom surface of the substrate and the ambient, respectively. R_{g-c}, R_{c-s} and R_{cool} denote the thermal resistances from cover glass to the cell junction,

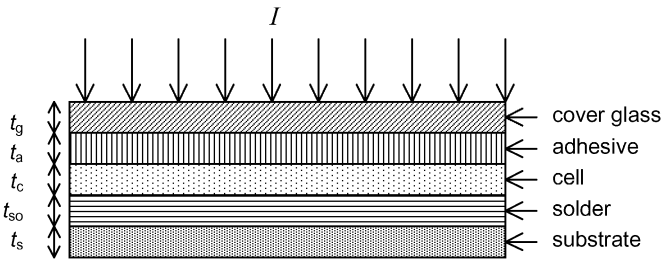


Fig. 4. Cell and mounting layers with thicknesses t.

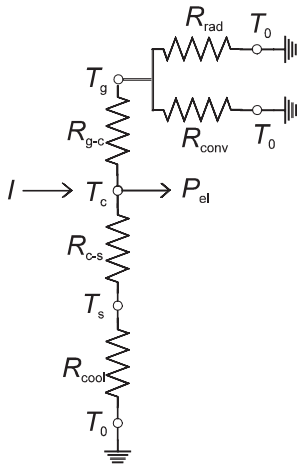


Fig. 5. Equivalent thermal circuit of cell, mounting and cooling system.

from cell junction to substrate bottom, and from substrate, through the cooling system, to the ambient. T_c denotes the temperature of the cell junction, which is assumed to be in the middle of the cell. This temperature determines the efficiency of the cell. The simple model assumes that all incoming radiation is transmitted through the cell encapsulants and absorbed at the cell junction, where a percentage determined by the cell temperature is converted to electricity, and the remainder is converted to heat. It is also assumed that some heat is lost through radiation and convection from the cover glass surface, and that the remainder of the heat is removed by the cooling system on the substrate surface.

2.1. Heat loss through radiation and natural convection

The radiative heat flux (per unit area) is related to the cover glass surface temperature as follows [19]:

$$\dot{q}_{rad} = \varepsilon\sigma(T_g^4 - T_0^4), \tag{3}$$

where T_g is the surface temperature, T_0 is the ambient temperature, ε is the surface emissivity and σ is the Stephan–Boltzmann constant. However, for simplification, it is common to linearise this equation in the following manner [2]:

$$\dot{q}_{rad} = 4\varepsilon\sigma T_0^3(T_g - T_0). \tag{4}$$

For an ambient temperature of 25 °C, this approximation gives an error in $\dot{q}_{rad}$ of less than 5% for cell temperatures up to 170 °C.

By determining a thermal resistance R_{conv} for convective heat transfer from a surface, depending on surface and ambient parameters, the heat flux through convection from the surface is simply given by

$$\dot{q}_{conv} = \frac{T_g - T_0}{R_{conv}}. \tag{5}$$

2.2. Electrical power output

The cell efficiency varies with both temperature and concentration. There are various models for temperature and concentration dependency found in literature [2,3,18,20,21]. As shown in Fig. 6, most of the models predict quite similar dependencies in the lower temperature range; most models assume straight lines. The different values predicted arise from the fact that cells have different peak efficiencies. Therefore, a simple approach is used in this article by assuming a linear decrease in efficiency with temperature, and no dependency on concentration, as in Ref. [20]. This gives the following model:

$$\eta = a(1 - bT_c), \tag{6}$$

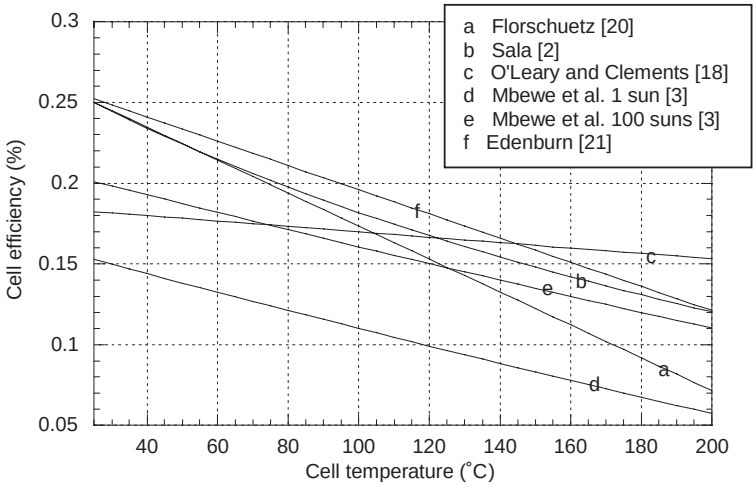


Fig. 6. Comparison of different models for cell efficiencies at various temperatures.

where a and b are parameters describing a particular cell (for illustrative purposes, we use the values given in Ref. [20]), and η is the cell efficiency at a given cell temperature T_c. The electrical output is given by

$$P_{el} = \eta T_c. \tag{7}$$

2.3. Energy balance

If I denotes the incoming concentrated solar flux, and

$$\dot{q}_{cool} = \frac{T_s - T_0}{R_{cool}} \tag{8}$$

is the thermal energy removed by the cooling system, the following relation must be satisfied to achieve thermal equilibrium:

$$I - \dot{q}_{rad} - \dot{q}_{conv} - P_{el} - \dot{q}_{cool} = 0. \tag{9}$$

Solving Eqs. (4)–(9) gives the value for T_c at any given illumination value. It should be noted that $\dot{q}_{cool}$ is very large compared to $\dot{q}_{rad}$ and $\dot{q}_{conv}$ in most cases of concentration, and so the significance of the model and parameters chosen for these aspects of the actual cells becomes less important.

Fig. 7 shows the electrical power output that would result from various illumination levels using this model and the values given in Table 1. The different curves correspond to different values of R_{cool}. There is clearly a definitive maximum power output for all curves. However, these curves must be seen together with Fig. 8,

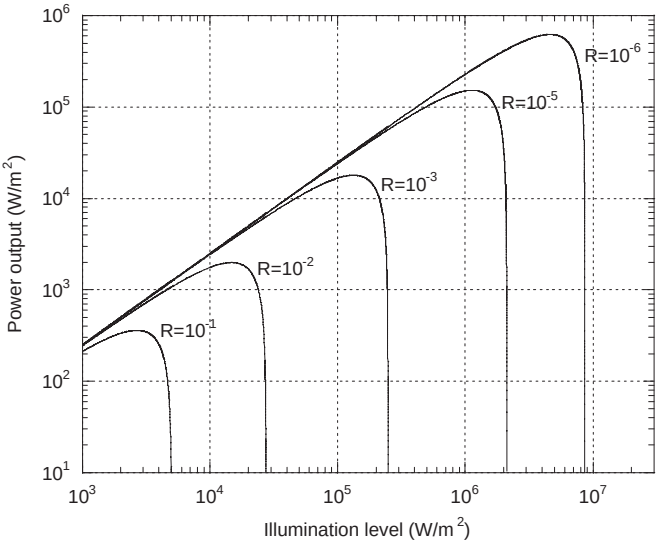


Fig. 7. Electrical power output versus illumination level for various values of R_{cool} (K m^2/W).

Table 1
Parameters used in thermal model

Layer	Material	Thickness t (m)	Thermal	conductivity k (W/m K)	Total thermal resistance $R = \sum_i \frac{t_i}{k_i}$ (K m^2/W)
Cover glass	Ceria-doped glass [5]	3×10^{-3}	1.4 [19]		
Adhesive	Optical grade RTV (room temperature vulcanization) silicone [5]	1×10^{-4}	145 [2]		
Top half of cell	Silicon [5]	6×10^{-5} [5]	145 [2]		$R_{g-c} = 2.14 \times 10^{-3}$
Bottom half of cell	Silicon [5]	6×10^{-5} [5]	145 [2]		
Solder	Sn:Pb:As: [2]	1×10^{-4} [2]	50 [2]		
Substrate	Aluminum nitride [5]	2×10^{-3} [2]	120 [5]		$R_{c-s} = 1.91 \times 10^{-5}$

Other parameters Symbol	Description	Value	Symbol	Description	Value
T_0	Ambient temperature	25 °C	R_{conv}	Convective thermal resistance	0.2 K m^2/W [2]
ε	Hemispherical surface emissivity	0.855 [22]	a	Cell efficiency parameter	0.5546 [20]
σ	Stephan–Boltzmann constant	5.67×10^{-8} W/m^2 K^{-4} [19]	b	Cell efficiency parameter	1.84×10^{-4} K^{-1} [20]

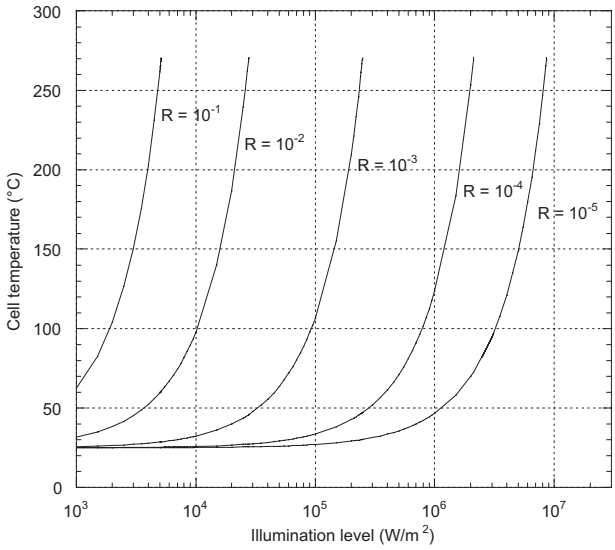


Fig. 8. Cell temperature versus illumination level for various values of R_{cool} (K m^2/W).

which shows the cell temperature rise with increasing concentration. It shows that the maximum power points correspond with very high cell temperatures. The actual power output will be limited by the bounds on the cell operating temperature. This implies temperature is always the limiting factor for concentrator cells. A low thermal resistance in the cooling system is crucial, and becomes even more important with increasing concentration level.

Fig. 8 clearly shows the thermal resistance bounds on various illumination levels. If cell temperatures are to be kept below 60 °C, and an insolation level of 1 kW/m^2 is assumed, then a thermal resistance of 10^{-3} K m^2/W would be feasible for concentrations up to 20 sun, while 10^{-5} K m^2/W is needed at 1000 sun. It should be noted that because of nonuniform flux distributions over the receiver surface, the peak flux is generally much higher than the mean concentration level, and the cooling system should be designed with peak intensities in mind.

3. Examples of cooling of concentrating PV in literature

In the textbook *Cells and Optics for Photovoltaic Concentration,* edited by Luque, there is an informative chapter by Sala on the cooling of solar cells [2]. It does not focus on concentrating PV in particular. The text presents models for calculating heat transfer through cells and the temperature effect on solar cell parameters. It also contains separate discussions on passive cooling through radiation, natural convection and conduction, and on forced liquid cooling. The text has been widely used as a reference for other research dealing with photovoltaic cooling systems.

Florschuetz [20] presents another general, theoretical approach to the cooling of solar cells under concentration. He uses the relations between illumination, cell temperature and cell efficiency to find an equation for the illumination level that gives the maximum power output for a given cooling system. This would be the equivalent of the equation for a line passing through the peaks in Fig. 7. However, as explained earlier, the maximum power points coincide with very high cell temperatures. The possibility of cell degradation has not been taken into account in this model. Florschuetz also explores the importance of contact resistance between the cell and the cooling system (represented by R_{c-s} in Section 2.1). He shows that the relative importance of the contact resistance increases substantially as the illumination levels rise. This is because the temperature difference across a boundary is given by $\Delta T = \dot{q} \cdot R$ and thus it increases with increasing heat flux $\dot{q}$ and increasing thermal contact resistance R. In high-concentration systems where $\dot{q}$ is large, a small contact resistance is needed to achieve the same temperature difference.

3.1. Single cell geometry

As described in the following section, passive cooling is found to work well for single-cell geometries for flux levels as high as 1000 sun. This is because of the large area available for heat sinking, as described in Section 1.2.1.

3.1.1. Passive cooling

Edenburn [21] performs a cost-efficiency analysis of a point-focus Fresnel lens array under passive cooling. The cooling device is made up of linear fins on all available heat sink surfaces (see Fig. 9). Concentration values under consideration are 50, 92 and 170 sun. The analysis consists of using given values for the cost of aperture (lens and cell) area and for cooling device area and cost optimising the cooling geometry. Cell degradation at high temperatures is not considered. This implies that arrays that employ the passive cooling devices developed under this model must have a mechanism for defocusing under extreme thermal conditions (very low wind speed, high insolation and high ambient temperatures). In the search for cost-effectiveness, Edenburn also suggests housing the cell assembly in a painted aluminum box, and to use the bottom of this as a finless heat sink. He states that

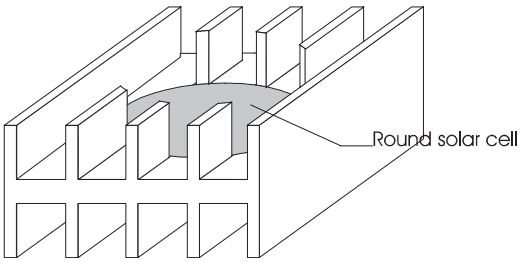


Fig. 9. Passive heat sink for a single cell as suggested by Edenburn [21].

during calm air conditions, radiation is the most important component of heat loss. A finned surface will radiate less than a finless one because of the temperature drop from the base of the fins to the tips. With this design, the cells could probably be kept below 150 °C even on extreme days at a concentration level of about 90 suns. Edenburn concludes that for point focus arrays, the cost of passive cooling increases with lens area, while it remains almost unchanged with concentration. The reason is that as the aperture area is increased, a thicker and more expensive heat exchanger is required. When concentration level increases, the heat sink optimal design does not change by much, but a low contact thermal resistance between the substrate and the heat sink becomes increasingly important to keep the cell temperature down.

Miñano [16] presents a thermal model for the passive cooling of a single cell under high concentrations. Like Edenburn, he concludes that passive cooling is increasingly efficient for cells as their size is reduced. Comparing the given cell efficiencies of the GaAs cells used in this case, it seems likely that a concentration of 1000 sun would be possible as long as the temperatures are kept low. Miñano advises that cells be kept below 5 mm diameter. Heat sinks for these cells would be similar to those used for power semiconductor devices.

Araki et al. [17] presents further results that show the effectiveness of passive cooling of single cells. In this study, an array of Fresnel lenses focus the light onto single cells mounted with a thin sheet of thermally conductive epoxy onto a heat-spreading aluminum plate. The concentration level is about 500 suns. Outdoor experiments show a temperature rise of cells over ambient of only 18 °C, without conventional heat sinks. It is shown that good thermal contact between the cell and the heat spreading plate is crucial to keep the cell temperature low. Techniques to enhance this could be to use a thinner epoxy layer, or to increase the thermal conductance of the epoxy.

Graven et al. [23] have patented a single cell lens array which employs a heat sink with longitudinal fins. The thermal contact between the cells and the heat sink is provided by a set of rods with springs that force the surfaces together. A thin polyester film between the cells and the heat sink ensures both good thermal contact and electrical insulation.

3.1.2. Active cooling

Edenburn [21] also considers using active cooling on his point focus arrays described above. Cells are placed in rows with one rectangular coolant channel run along the back of each row. To enable a cost comparison between the different cooling regimes, the possible advantage of using the extracted heat for thermal energy supply purposes is not taken into consideration. However, Edenburn concludes that if this were done, active cooling would almost certainly be the most cost-efficient solution. Without this extra advantage, however, the parasitic power losses involved in pumping and in dissipating the waste heat make active cooling more expensive than passive cooling for single cells. The only exemption would be for very large lenses (more than 30 cm in diameter). At this size, the costs of active and passive cooling become almost the same [21].

3.2. Linear geometries

3.2.1. Passive cooling

Florschuetz [24] uses his model to assess both active and passive cooling options for a linear geometry. For the passive solution, cells are mounted along either a planar or a finned metal strip. The illumination levels at the maximum power points (see above) are compared for the different cooling systems. Pin fins are found to perform better than plane ones, but because pin fins are more costly to manufacture, they may still not be the best option. The model suggests that the plane strip would be sufficient for very low concentration levels (less than 5 suns) and the finned strip only for slightly higher levels (10 suns). With 2.2 m/s wind speed, the plane strip should work up to about 10 suns and the finned one up to 14 suns. Note that this analysis does not take cell efficiency degradation into account.

The EUCLIDES is a trough-type photovoltaic concentrator technology originating from Spain [25]. In this system, thermal energy is passively transferred to the ambient through a lightweight aluminium-finned heat sink. The fins have been optimised for the relatively low concentration (about 30 suns) used on the EUCLIDES system. The optimisation gave fin dimensions to be 1 mm thick, 140 mm long and spaced about 10 mm apart. This could not be manufactured by ordinary means, but was accomplished by stacking fin- and separator-plates, and tightening them with screws. This method is quite costly. The heat sink is projected to contribute to 15.7% of the total cost of an EUCLIDES-type plant, while photovoltaic modules and the mirrors contribute 11.9% and 10.8%, respectively. The operating cell temperature has been measured to be about 58 °C.

Edenburn [21] considers the cooling of a linear trough design. In his system, cells are mounted in two lines in a V-type geometry. The passive heat exchanger consists of a finned mast that avoids shading the concentrator (Fig. 10). The concentration levels under consideration are 20, 30 and 40 suns. Edenburn finds that because of higher cell temperatures, resulting from the longer path length for the heat to be conducted to the fins of the heat sink, passive cooling of a linear design is much more expensive than for a single cell design, and it does not seem to be cost-efficient for this setup. To increase the performance, he suggests filling the cavity of the 'mast' with an evaporative fluid that would work as a thermosyphon to transport heat away from the cells at a very low temperature differential.

The heat pipe approach is further explored by Feldman et al. [26] on a concentration ratio of about 24 suns. The 'mast' is made out of extruded surface aluminium, and the evaporative working fluid is benzene. This gives a maximum evaporator surface temperature of about 140 °C. The cell temperature would be even higher than this given the thermal resistance between the cell and the evaporator surface. The model shows that the heat transfer in this system is highly dependent on the condenser surface area. The prototype has an evaporator area of 0.61 m^2 and a condenser area of 2.14 m^2. Outdoor testing also shows that the operating temperature is a strong function of wind speed, and less of ambient temperature, wind direction and mast tilt angle. Under the worst case scenario, which is an ambient temperature of 40 °C and 19.2 kW/m^2 illumination, a minimum wind speed

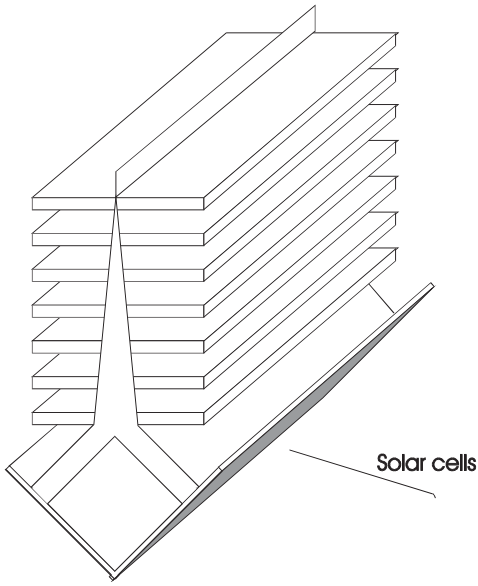


Fig. 10. Passive cooling of a linear design as suggested by Edenburn [21].

of 1 m/s is required to keep the evaporator temperature below 140 °C. The surface area would have to be increased by a factor of 2.1 to achieve the same in no-wind conditions. Thermal resistance from base surface to the ambient is 0.114 K m^2/W in the 1 m/s wind case.

Akbarzadeh and Wadowski [27] report on a linear, trough-like system which also uses heat pipes for cooling. In this case, the reflector is not a parabola, but an 'ideal reflector' which is said to give a uniform illumination across the cells. Each cell is mounted vertically on the end of a thermosyphon, which is made of a flattened copper pipe with a finned condenser area (Fig. 11). The system is designed for 20 suns concentration, and the cell temperature is reported not to rise above 46 °C on a sunny day, as opposed to 84 °C in the same conditions but without fluid in the cooling system.

3.2.2. *Active cooling*

Florschuetz [24] considers cooling his strip of cells actively by either forced air through multiple passages or water flow through a single passage. He notes that with forced air cooling, there is a substantial temperature rise along the cells due to the low heat capacity of air. The required pumping power is also quite large compared to the effective cooling. For these reasons, forced air cooling does not seem to be a viable alternative. Water cooling, on the other hand, permits operation at much higher concentration levels.

Edenburn [21] suggests a cooling system for his linear design that consists of a channel of quadratic cross-section, tilted 45°, with the V-shaped PV receiver placed

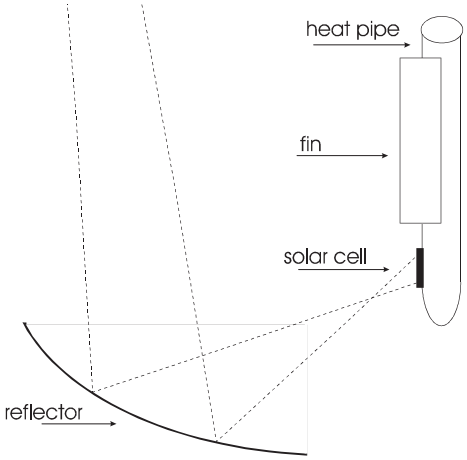


Fig. 11. Schematic of heat pipe based cooling system as suggested by Akbarzadeh and Wadowski [27].

on two of the channel sides. Active cooling was found to be more cost-efficient than passive cooling in linear designs.

O'Leary and Clements [18] give a theoretical analysis of the thermal and electrical performance of an actively cooled system. The cooling methods considered consist of various geometries of coolant flow through extruded channels, the coolant liquid being water–ethylene glycol mixture. An optimal geometry is suggested based on maximum net collector output versus coolant flow. The required pumping power rises proportionally with increased coolant mass flow rate, which is characteristic for laminar flow in channels. Although it would seem favourable to operate at the highest possible mass flow rate in order to obtain the lowest cell temperatures and highest cell performances, there is actually shown to be a definite optimum operation region, because the rate of increase in R drops as the mass flow increases.

A system of linear Fresnel lenses, cooled by water flow through a galvanised steel pipe, is described by Chenlo and Cid [11]. The system has a concentration level of about 24 sun. The cells are soft soldered to a copper–aluminum–copper sandwich, which is in turn soldered to the rectangular pipe. This mounting gives a satisfactory cell to steel tube thermal resistance ($R = 8 \times 10^{-5}\,\mathrm{K\,m^2/W}$). The soft soldering allows for some difference in the thermal expansions between the cells and the steel tube to be accommodated. The convective thermal resistance of the coolant tube is found to be $R = 8.7 \times 10^{-4}\,\mathrm{K\,m^2/W}$ for Reynolds number $Re = 5000$. This paper also presents good electrical and thermal models for uniform and non-uniform cell illuminations.

Russell [28] has patented a heat pipe cooling system. His design uses linear Fresnel lenses, each focusing the light onto a string of cells mounted along the length of a heat pipe of circular cross-section (Fig. 12). Several pipes are mounted next to each other to form a panel. The heat pipe has an internal wick that pulls the liquid up to the heated surface. Thermal energy is extracted from the heat pipe by an internal coolant circuit, where inlet and outlet is on the same pipe end, ensuring a uniform temperature along the pipe. The coolant water is fed and extracted by common

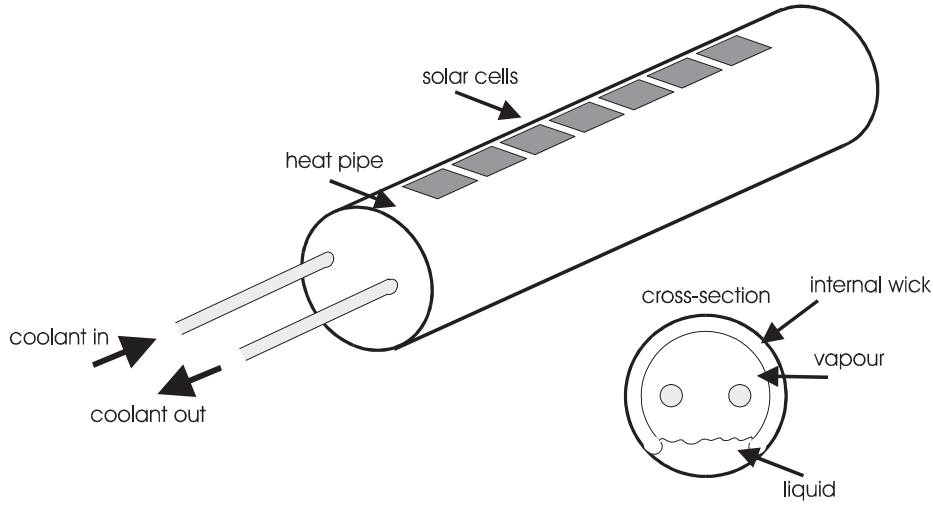


Fig. 12. Heat pipe based cooling system as suggested by Russell [28].

distribution pipes. An alternative system where the coolant enters at one end of the pipe and leaves at the other is also considered, but found to be less preferable because this would cause a substantial temperature gradient along the pipe length. Nothing is reported about the concentration level of the system. It is estimated to be low, because of the inherent limitations on heat pipes, which suffer from burnout at low operating temperatures (see Section 4.1).

The CHAPS system at the Australian National University [29] is a linear trough system where the row of cells is cooled by liquid flow through an internally finned aluminum pipe. The coolant liquid is water with anti-freeze and anti-corrosive additives and the optical concentration is $37\times$. Under typical operating conditions (fluid temperature 65 °C, ambient 25 °C, direct insolation $1\,kW/m^2$), the thermal efficiency is 57% and the electrical efficiency is 11% for the prototype collector. The cells, which are manufactured at ANU, are run at a fairly high temperature. Nothing is reported about the temperature gradient along the line of cells, which would result from the single coolant pipe, and whether this has a significant result on cell performance. This may be because the preliminary results are from a shorter prototype collector where the temperature difference is insignificant.

3.3. Densely packed cells

No reports of passive cooling of densely packed cells under concentration have been found.

3.3.1. Active cooling

Verlinden et al. [30] describe a monolithic silicon concentrator module with a fully integrated water-cooled cold plate. The module consists of 10 cells and is supposed

to act as a 'tile' in a larger array. With an optimised coolant flow rate of 0.0127 kg/s on an area of 36 cm^2, the total thermal resistance from cell to water (including all layers in between) is measured to be 2.3×10^{-4} K m^2/W. The design is further described by Tilford et al. [31], with module pictures and some further specifications. However, details are not given on the way in which the water flows through the cold plate.

Lasich [32] recently patented a water cooling circuit for densely packed solar cells under high concentration. The circuit is said to be able to extract up to 500 kW/m^2 from the photovoltaic cells, and to keep the cell temperature at around 40°C for normal operating conditions. This concept is based on water flow through small, parallel channels in thermal contact with the cells. The cooling circuit also forms part of the supporting structure of the photovoltaic receiver. It is built up in a modular manner for ease of maintenance, and provides good solutions for the problem of different thermal expansion coefficients of the various materials involved.

Solar Systems Pty. Ltd. has reported some significant results from their parabolic dish photovoltaic systems located in White Cliffs, Australia [13,33]. They work with a concentration of about 340 sun, and use the above-mentioned patent [32] for cooling the cells. With a water flow rate of 0.56 kg/s over an area of 576 cm^2 and an electrical pumping power of 86 W, they maintain an average cell temperature of 38.52 °C and achieve a cell efficiency of 24.0% using the HEDA312 Point-Contact solar cells from SunPower [33]. If all of the thermal energy extracted were being used, the overall useful energy efficiency in this system would be more than 70%. This demonstrates clearly the benefits of active cooling if one can find uses for the waste heat.

Vincenzi et al. [34,35] at the University of Ferrara have suggested using micromachined silicon heat sinks for their concentrator system. The photovoltaic receiver at Ferrara is 30×30 cm^2 and operates at a concentration level of 120 sun. By using a silicon wafer with microchannels circulating water directly underneath the cells, the cooling function is integrated in the cell manufacturing process. Microchannel heat sinks will be presented in more detail in Section 4.3.1. The reported thermal resistance is 4×10^{-5} K m^2/W, which is comparable to other microchannel systems (see Table 2), although perhaps slightly higher.

A system is patented by Horne [6] in which a paraboloidal dish focuses the light onto cells mounted in quite an innovative way. Instead of being mounted on a horizontal surface, they are situated vertically on a set of rings, designed to cover all of the solar receiving area without shading. Water is transported up to the receiver by a central pipe and then flows behind the cells, cooling them, before running back down through a glass 'shell' between the concentrator and the cells (Fig. 13). In this way, the water not only cools the cells, it also acts as a filter by absorbing a significant amount of UV radiation that would otherwise have reached the cells. Normally, cells need to be protected from UV radiation by a cover glass or lenses. In Horne's case, the water also absorbs some of the low-energy radiation, resulting in higher cell efficiency and a lower amount of power converted to heat in the cells. The patent incorporates a phase-change material in thermal contact with the cells, which works to prevent cell damage at "worst-case scenario" high temperatures.

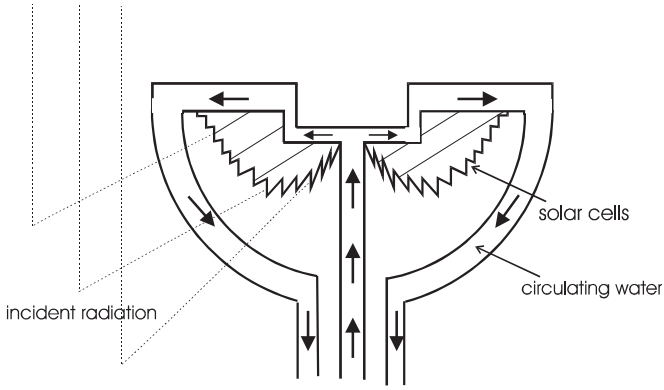


Fig. 13. Cooling of dense module as suggested by Horne [6].

An idea somewhat similar to that of Horne is patented by Koehler [36]. His idea is to submerge the cells in a circulating coolant liquid, whereby heat is transferred from two cell surfaces instead of just one. In this way the coolant also acts as a filter by absorbing much of the incoming low-energy radiation before it reaches the cells. The coolant liquid must be dielectric in order to provide electrical insulation of the cells. By choosing the right coolant fluid and pressure, one can achieve local boiling on the PV cells, which give a uniform temperature across the surface and a much higher heat transfer coefficient.

4. Other cooling applications

Cooling problems are not exclusive to photovoltaics. Recently, extensive research has been performed on the issue of cooling of electronic devices. The rapid progress towards denser and more powerful semiconductor components require the removal of a large amount of heat from a confined space [37–42]. Other areas where much research is being conducted on the subject of cooling include the nuclear energy and gas turbine industries. Both of these have a large cooling load and strict temperature limitations due to material properties. These applications generally deal with larger areas and different geometries from the electronics industry. Research from these three fields should provide a broad base for finding better options for cooling of photovoltaics.

The following section presents some studies that might be relevant for PV cooling, especially for the more demanding cases like densely packed cells under high concentration. Where figures are included these are generally the lowest thermal resistances reported in the studies. These provide some opportunity for comparison but it should be noted that they correspond to a wide range of flow rates, pumping powers, pressure drops and geometries. The lowest thermal resistance found in a study is often limited by the experimental equipment available. Thus, caution should be used when comparing these numbers.

4.1. Passive systems

There is a wide variety of passive cooling options available. The simplest ones involve solids of high thermal conductivity, like aluminum or copper, and an array of fins or other extruded surface to suit the application. More complex systems involve phase changes and various methods for natural circulation.

It should be noted that passive cooling is just a means of transporting heat from where it is generated (in the PV cells) to where it can be dissipated (the ambient). Complex passive systems reduce the temperature difference between the cells and the ambient, or they can allow a greater distance between the cells and the dissipation area. However, if the area available for heat spreading is small and shading is an issue, no complex solutions will help avoid the use of active cooling. Heat dissipation is still limited by the contact point between the terminal heat sink and the ambient, where the convective heat transfer coefficient, and less so the radiative heat transfer (except at very high temperatures), are the limiting factors.

Kraus and Bar-Cohen [43] give an extensive and very useful introduction to the design of heat sinks. Their book contains an overview of typical convective thermal resistances for different configurations, as a useful guide when choosing the cooling system. It also presents a step-by-step procedure for heat sink design and optimisation procedures both for single fins and fin arrays. Optimum dimensions for fins of common heat sink materials are given, as well as the heat transfer properties for optimised arrays.

One way of passively enhancing heat conduction is the use of heat pipes. The theory on and use of these devices is thoroughly described by Dunn and Reay [44]. The use of heat pipes is not feasible for high concentrations because heat pipe performance is limited by the working fluid saturation temperature and the point at which all liquid evaporates (burnout). With water, a heat flux of 250–1000 kW/m^2 can be accommodated but only at temperatures above 140 °C. In the search for better cooling options for computer components, heat pipes provide an alternative for transporting the heat away from the component and to a place better-suited for a fan or other heat sink (remote heat exchangers). Pastukhov et al. [45] and Kim et al. [41] show promising results for these systems. Launay et al. [46] study the effect of microheat pipe arrays etched into the silicon wafer. They show an improvement of conductivity through the silicon, depending on the geometry of the heat pipes and the fluid charge. Xuan et al. [47] describe the flat plate heat pipe (FPHP), which is a flat copper shell filled with a working fluid. A layer of sintered copper powder is applied to the heated surface of the FPHP in order to enhance heat transfer. The FPHP is studied under various orientations. When installed horizontally, the extra working fluid forms a liquid layer on the heated surface and reduces heat transfer. The best result is achieved when the FPHP is installed in the vertical direction, when the working fluid is distributed across the heated surface by the capillary action of the sintered layer, ensuring there is not too much fluid at the surface at any time. It is shown that the FPHP is a good alternative to a solid heat sink due to its low thermal resistance, isothermality and lightweight features.

Chen and Lin [48] study the capillary pumped loop used as a heat transfer device. Their system is capable of dissipating a heat load of $25\,kW/m^2$ from an area of $4.2 \times 3.8\,cm^2$ while keeping the heated surface below $100\,^{\circ}C$. The device performs best if the vertical distance between the evaporator and the condenser is higher than 1 cm. The effect of orientation is not included in the study.

4.2. Forced air cooling

The thermal properties of air make it far less efficient as a coolant medium than water [43]. This implies that more parasitic power (to power fans) will be needed to achieve the same cooling performance. Air cooling is also in general less suited to the secondary use of thermal energy from the PV absorbers. Hence, air is a less favourable option in many cases. However, in some situations where water is limited, forced air may still be the preferred option. The heat transfer of forced air cooling can be enhanced in much the same ways as with water. Detailed information on the design of forced air heat sinks can be found in Ref. [43]. Other studies on forced air cooling are not included in this review.

4.3. Liquid single-phase forced convection cooling

4.3.1. Microchannel heat sinks

The microchannel heat sink is a concept well-suited to many electronic applications because of its ability to remove a large amount of heat from a small area. Tuckerman and Pease [49] were the pioneers who first suggested the microchannel heat sink, based on the fact that the convective heat transfer coefficient scales inversely with the channel width. The best reported thermal resistance from their experiments was $9.0 \times 10^{-6}\,K\,m^2/W$ for a heated area and heat sink of $1 \times 1\,cm^2$, flow rate of 8.6 ml/s and a pressure drop of 213.7 kPa. This significantly raised the experimental limit on heat removal per area, and may have allowed for further miniaturisation of electronic components [49].

Later studies have showed two major drawbacks to the microchannel heat sink. These are a large temperature gradient in the streamwise direction, and a significant pressure drop that leads to high pumping power requirements. Much work has been published on the modeling and optimisation of various aspects of the microchannel heat sink [40].

Ryu et al. [38] presents a numerical optimization that minimises the thermal resistance subject to a specified pumping power. For a heat sink of $1 \times 1\,cm^2$, the lowest reported thermal resistance is $9 \times 10^{-6}\,K\,m^2/W$. The associated pressure drop is 103.42 kPa and the optimal dimensions are 56 μm channel width, 44 μm wall width, and 320 μm channel depth. More results and discussion on the pressure drop and heat transfer in a heat sink of rectangular microchannels is given by Qu and Mudawar [50]. Their modelling and experiments deal with laminar flow only. Channel dimensions were 231 μm width and 713 μm depth. Qu and Mudawar conclude that conventional Navier–Stokes and energy conservation equations can

accurately predict the pressure drop and heat transfer characteristics for microchannels of these dimensions.

An experimental study of heat transfer in rectangular microchannels by Harms et al. [51] concludes that heat transfer performance can be increased by decreasing the channel width and increasing the channel depth. Developing laminar flow is found to perform better than turbulent flow due to the larger pressure drop associated with turbulent flow. The lowest reported thermal resistance is $1.26 \times 10^{-4}\,\mathrm{K\,m^2/W}$ for a flow rate of 118 ml/s over an area of $39.3\,\mathrm{cm^2}$ and a 169 kPa pressure drop.

Owhaib and Palm [52] present an experimental study which verifies the best correlations to use for modelling heat transfer in circular microchannels. Tubes of three different diameters were studied. The results show that in the laminar flow regime, the heat transfer coefficient is largely independent of channel diameter, while in the turbulent regime ($Re > 6000$), smaller channels are clearly better. The best reported thermal resistances are $10^{-4}\,\mathrm{K\,m^2/W}$ for 0.8 mm tubes in the turbulent flow regime, and $4 \times 10^{-4}\,\mathrm{K\,m^2/W}$ for laminar flow. No data on pressure drops or flow rates are given.

The effect on tip clearance on the thermal performance of microchannels has also been studied. Tip clearance denotes the spacing between the channel walls and the top surface. It has generally been assumed that tip clearance would lower the efficiency of the heat sink because of the phenomenon of flow bypass: as the tip clearance is raised, for a given pumping power, the flow rate will decrease between the channels while increasing through the tip clearance. As a result, less heat is transferred near the base of the channels. However, Min et al. [53] found that in microchannel heat sink, the added heat transfer through the fin tips lead to an increased heat sink performance as long as the ratio of tip clearance to channel width is kept below 0.6. Similar results are found by Moores and Joshi [54] for a shrouded pin fin heat sink.

The search for a microchannel design that deals with the problem of non-uniform temperatures and pressure drops has been carried out by a number of researchers, and several innovative solutions have been found. Alternating flow directions is one way of reducing the streamwise temperature gradient in the microchannel heat sink. The single layer counter flow technique was proposed by Missagia and Walpole [55]. Their design consists of a silicon wafer with microchannels machined into them, attached to a manifold plate that directs the water to flow in alternating directions through the channels. The results indicate a thermal resistance of $1.1 \times 10^{-5}\,\mathrm{K\,m^2/W}$, for a laminar flow of 28 ml/s. The associated pressure drop for a 10 cm long heat sink would be 452 kPa. Vafai and Zhu [37] suggest using two layers of counter-flow microchannels. Numerical results show that the streamwise temperature gradient is significantly lowered compared to a one-layer structure. This in turn allows for a smaller pressure drop to fulfil the same cooling requirements. No specific data for thermal resistances or pressure drops are given.

Chong et al. [40] optimised the counter flow principle for single and double layer channels of the designs described above. The simulations model both designs for laminar and turbulent flows. The results show that laminar flow is to be preferred over turbulent for both cases. The single layer counter flow heat sink gives an overall

thermal resistance of $6.9 \times 10^{-6}\,\mathrm{K\,m^2/W}$ with a pressure drop of 122.4 kPa. For the double layer design the values are $6.6 \times 10^{-6}\,\mathrm{K\,m^2/W}$ and 54.6 kPa, both under laminar flow conditions. The paper does not arrive at any conclusions as to whether single or double layer counter flow is the preferable alternative.

A two-layered microchannel heat sink with counter flow, called the manifold microchannel heat sink, is also designed to lower the temperature gradient and pressure drop. This design has been modelled and optimised by Ryu et al. [42]. In the manifold microchannel heat sink, the coolant flows through alternating inlet and outlet manifolds in a direction normal to the heat sink (Fig. 14). This way the fluid spends a relatively short time in contact with the base, thus resulting in a more uniform temperature distribution across the surface to be cooled. With laminar flow, it is shown that the thermal resistance is lowered by more than 50% compared to the traditional microchannel heat sink, while drastically reducing the temperature variations on the base. A number of numerical calculations are performed to find the optimal channel depth, channel width, fin thickness and inlet/outlet width ratios. All optimisations are constrained by a given pumping power. Optimal dimensions are found to be divider width $\geqslant 500\,\mu\mathrm{m}$ and inlet width + outlet width $\geqslant 1000\,\mu\mathrm{m}$, with an associated thermal resistance of $3.1 \times 10^{-6}\,\mathrm{K\,m^2/W}$.

Inspired by the superior mass flow capacity of the mammalian circulatory and respiratory system, Chen and Cheng [56] use this idea to design a fractal net of microchannels. On a purely theoretical basis, they conclude that fractal-like microchannels can increase the heat transfer while reducing the pressure drop when

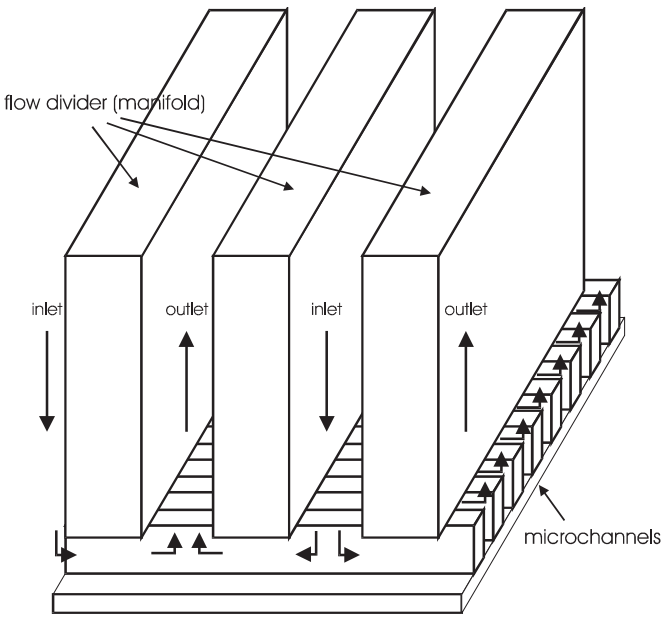


Fig. 14. Manifold microchannels as suggested by Ryu et al. [42].

compared with parallel microchannels. This is based on the assumptions of laminar, fully developed flow, and negligible pressure drop due to bifurcation.

4.3.2. Impinging jets

Very low thermal resistances (generally 10^{-5}–$10^{-6}\,K\,m^2/W$) [57] can be achieved through the use of impinging liquid jets. When high velocity liquid is forced through a narrow hole (axisymmetric jet) or slot (planar jet) into the surrounding air, and onto a surface to be cooled, a free surface forms. The impinging jets are capable of extracting a large amount of heat because of the very thin thermal boundary layer that is formed in the stagnation zone directly under the impingement, and that extends radially outwards from the jet. However, the heat transfer coefficient decreases rapidly with distance from the jet. To cool larger surfaces, it is therefore desirable to use an array of jets. A problem arises when water from one jet meets the water from the neighbouring jet. Disturbances arise which are difficult to model accurately but have been shown to decrease the overall heat transfer drastically [58,59]. If measures are taken to deal with this 'spent flow' (through drainage openings), impinging jets are predicted to be a superior alternative to microchannel cooling [59] for target dimensions larger than the order of $0.07 \times 0.07\,m^2$.

Webb and Ma [58] give an extensive overview of the literature available on liquid impinging jets. Their review distinguishes between free and submerged jets, and axisymmetric and planar jets, and deals with single-phase jets only. The article points out a number of areas where further studies are needed. These include the effect of curved surfaces and spent flow, and the local heat transfer coefficient at points other than the stagnation zone directly underneath the jet.

Womac et al. [60] present an experimental study of the heat transfer coefficient in free and submerged 2×2 and 3×3 arrays of liquid jets without treatment of spent flow. The effect of nozzle-to-plate spacing is studied, and found to be insignificant for free jets, but to have an effect on submerged jets. Correlations for the heat transfer in both types of jets are presented.

4.4. Two-phase forced convection cooling

By allowing the coolant fluid to boil, the latent heat capacity of the fluid can accommodate a significantly larger heat flux and achieve an almost isothermal surface. Although any comprehensive heat transfer textbook such as [19] will give an introduction to forced convection boiling, two-phase flows are complicated to model.

When the bulk liquid is below saturation temperature, but the heat flux is high enough that liquid at the surface can reach saturation temperature, sub-cooled boiling occurs. Under sub-cooling, bubbles will collapse as they are released from the wall and travel into the surrounding liquid. Sub-cooled forced convection boiling in small channels is among the most efficient heat transfer methods available [61,62]. This is often used in applications with extremely large heat fluxes such as fusion reactors first walls and plasma limiters. The most important parameter in this case is the critical heat flux (CHF) defined as the point at which enough vapour is being

formed that the surface is no longer continuously wetted. If the heat flux is raised above the CHF, a very large increase in temperature will occur and most likely result in overheated and damaged equipment. Thus, to achieve maximum cooling, one wants to run the system close to the CHF, but never above. Higher heat transfer coefficients, and thus lower wall temperatures, can be found at lower heat fluxes. Predicting the CHF is difficult because it depends on a number of parameters. High velocities, large sub-coolings, small diameter channels and short heated lengths are known to increase the CHF.

Two-phase flows may be a good option for the cooling of photovoltaic cells when the heat fluxes are high. The saturation temperature of water can be brought to 50 °C at a pressure of 0.13 bar [19]. To avoid pressurised systems, other working fluids may be used, e.g. Vertrel XF [39].

A number of studies are devoted to the detailed analysis of bubble formation, onset of different boiling regimes, and CHF for subcooled boiling [61,63,64]. Bartel et al. [65] present a very good literature review on sub-cooled boiling. The review points out that there is a lack of available data on local measurements in the sub-cooled boiling region.

There are a number of studies dealing with two-phase flow in microchannels. Ghiaasiaan and Abdel-Khalik [62] give an extensive literature review of the subject. Microchannels with hydraulic diameters of the order 0.1–1 mm and long length-to-hydraulic diameter ratios are considered. Their review includes a thorough description of flow regimes in horizontal and vertical channels, correlations for pressure drops, forced flow subcooled boiling and CHF. Detailed studies of bubble formation and flow boiling in microchannels are also found in Ref. [66–68].

Hetsroni et al. [39] describes a microchannel heat sink that keeps the electronic device at a temperature of 50–60 °C, a temperature highly suited for photovoltaic purposes. The working fluid is Vertrel XF, which has the desired saturation temperature and is dielectric, so that it can be brought into contact with the active electronics. The study was performed at relatively low heat fluxes ($<60\,\mathrm{kW/m^2}$). Results show a much more uniform temperature across the surface compared to water cooling at comparable flow rates (temperature differences of 5 °C as opposed to 20 °C). However, some non-uniformities in heat transfer occurred because of two circumstances specific to parallel microchannels: the two phases may split unevenly on entering the channels, leading to different heat transfers for different channels; secondly, the wall superheat (the difference between the heated wall temperature and the liquid saturation temperature) for the onset of nucleate boiling is very low, something which leads to pressure fluctuations and uneven heat transfer. Temperature and pressure fluctuations are also found to be characteristic of boiling in minichannels by Hapke et al. [69]. The lowest thermal resistance reported by Hetsroni et al. was $9.5\times10^{-5}\,\mathrm{K\,m^2/W}$ at a mass flux of $290\,\mathrm{kg/m^2s}$.

Inoue et al. [70] study the use of boiling in confined jets to cool a very high heat flux (near $30\,\mathrm{MW/m^2}$) in a fusion reactor (see Fig. 15). This system proposes an innovative way of dealing with the spent flow, and at the same time preventing splash of water from the violent boiling that may occur at the surface under these conditions. The jets proposed are planar jets, but the experiments only look at the

two-dimensional version. Therefore, the potential problem of outgoing water heating the incoming water and thus lowering the cooling capacity is not considered. The CHF is studied as a function of jet flow velocity, sub-cooling and curvature of the heated surface. The results show that the CHF in confined flow is almost double that of a free flow jet. Surface curvature does not seem to cause any significant effect.

5. Comparison of cooling options

It is problematic to compare such a wide range of cooling options. Depending on the application, one may want to compare parameters such as pumping power, weight, materials use, ease of manufacturing and maintenance, maximum heat removal, temperature uniformity, shading, etc. All of these criteria are difficult to incorporate in a single analysis. In addition, most literature generally does not give information on all of these aspects.

Table 2 gives a summary of the various cooling options described in this review. In order to enable a comparison of pumping powers, which is an important parameter when it comes to power generating systems, the pumping power is estimated as $P = \dot{m} \times \Delta p$ [56] in cases where only mass flow rate and pressure drops are given. It should be noted that pressure drops might or might not incorporate manifolds or other external factors. Different analyses also use slightly different definitions for thermal resistances. Extra care should be taken when comparing different systems such as jets versus passive cooling or two-phase versus single-phase flows. Thermal resistances, flow rates and pumping powers are all given per unit area for easier comparison.

All precautions taken, Figs. 16–19 still provide an interesting comparison between options. The letters refer to the references given in Table 2. Results are from theoretical or experimental studies as indicated in the table. There is a wide variety between the different studies, even within the same categories (Fig. 16). This shows that experimental work is still very important for determining the best cooling methods. Fig. 17 shows how the required pumping powers vary over 5 orders of magnitude for similar thermal resistances. Figs. 18 and 19 show variations of almost four orders of magnitude for flow rates and pressure drops, respectively. A reason for these results may be that various studies are optimised with respect to different constraints. It would probably also depend on experimental limitations at the various facilities. The values cited are the lowest thermal resistances from each study.

What seems to perform best in all comparisons is the category ‘improved microchannels’ which includes various forms of alternating flow arrangements. This method provides the lowest thermal resistance along with low power requirements. In all microchannel studies, laminar flow seems to outperform turbulent. Etching microchannels into the silicon substrate as a part of the manufacturing process of photovoltaic modules may prove a very good option for photovoltaic cell cooling.

Impinging jets seem to be a promising alternative, provided measures are taken to deal with spent flow. No studies have yet come up with a solution to this problem when dealing with single-phase liquid flows.

Table 2
Values cited in references

Work	Configuration	Heated area (m^2)	Pump power (W/m^2)	Pressure drop (kPa)	Mass flow rate ($kg/m^2\,s$)	Thermal resistance $K\,m^2/W$	
Sala [2] (theoretical)	Air cooling, plane surface	—	—	—	—	2.0×10^{0b}	a
	Water cooling, plane surface: laminar mode	—	—	—	—	2.6×10^{-3}	b
	Turbulent mode	—	—	—	—	2.7×10^{-4}	c
Florshuetz [20] (theoretical)	No extruded surface, calm air	—			—	3.3×10^{-2b}	d
	Finned strip, calm air	—			—	1.1×10^{-2b}	e
	Forced air through multiple passages	1.52×10^{-1}			3.95×10^{-1}	2.6×10^{-3}	f
	Water cooling	1.52×10^{-1}			3.03×10^{0}	4.3×10^{-4}	g
	Impinging jet, nozzle—plate distance = 0.16 cm	2.58×10^{-3}			7.75×10^{0}	5.1×10^{-5}	h
Feldman et al. [26] (experimental)	Finned heat pipe, calm air	6.10×10^{-1}	—	—	—	9.8×10^{-3b}	i
Luque et al. [25] (experimental)	Finned strip, calm air	—	—	—	—	2.2×10^{-3b}	J
Chenlo and Cid [11] (experimental)	Water flow through rectangular steel pipe	—	—	—	—	8.7×10^{-4}	k
Coventry [29] (experimental)	Water flow through internally extruded channel	1.15×10^{-1}	—	—	3.48×10^{-1}	1.3×10^{-3}	L
Verlinden [30] (experimental)	Water cooled cold plate	3.60×10^{-3}	—	—	3.51×10^{0}	2.3×10^{-4}	m

Vincenzi et al. [35] (experimental)	Microchannels	3.40×10^{-5}	8.82×10^{2}	—	1.82×10^{1}	4.0×10^{-5}	n
Kraus and Bar-Cohen [43] (theoretical)	Parallel fin heat sink, calm air						
		1.68×10^{-2}	—	—	—	4.7×10^{-3}[b]	o
Tuckerman and Pease [49] (experimental)	Microchannels	1.00×10^{-4}	5.10×10^{5}	5.94×10^{3}	8.60×10^{1}	9.0×10^{-6}	p
Ryu et al. [38] (experimental)	Microchannels	1.00×10^{-4}	2.56×10^{4}	2.13×10^{2}	1.00×10^{2}	9×10^{-6}	q
Harms et al. [51] (theoretical)	Microchannels	3.93×10^{-3}	6.32×10^{3}[a]	1.69×10^{2}	3.74×10^{1}	1.3×10^{-4}	R
Owhaib and Palm [52] (experimental)	Circular microchannels, laminar flow						
	Turbulent flow	—	—	—	4.0×10^{-4}	S	
						1.0×10^{-4}	T
Missaggia and Walpole [55] (experimental)	Microchannels, single layer counter flow	2.30×10^{-4}	3.00×10^{4}[a]	2.48×10^{2}	1.21×10^{2}	1.1×10^{-5}	U
Chong et al. [40] (theoretical)	Microchannels, single layer counter flow, laminar	1.00×10^{-4}	7.70×10^{0}[a]	1.18×10^{2}	6.53×10^{-2}	6.9×10^{-6}	v
	Turbulent		5.04×10^{1}[a]	1.12×10^{2}	4.50×10^{-1}	4.8×10^{-6}	w
	Microchannels, double layer counter flow, laminar		5.25×10^{1}[a]	5.64×10^{2}	9.31×10^{-2}	6.6×10^{-6}	x
	Turbulent		1.48×10^{2}[a]	5.64×10^{2}	2.62×10^{-1}	5.8×10^{-6}	y
Ryu et al. [42] (theoretical)	Manifold microchannels	1.00×10^{-4}	1.50×10^{4}	—	1.40×10^{-1}	3.1×10^{-6}	z
Rohsenow et al. [57] (theoretical)	Impinging jets	—	—	—	—	1.0×10^{-6}	A
Hetsroni et al. [39] (experimental)	Two-phase microchannels	1.00×10^{-4}	8.70×10^{2}[a]	3.00×10^{0}	2.90×10^{2}	9.5×10^{-5}[b]	B

[a]Calculated from given data as $P = \dot{m} \times \Delta p$.
[b]Use caution with thermal resistances for natural convection or two-phase flow (see Section 1.3).

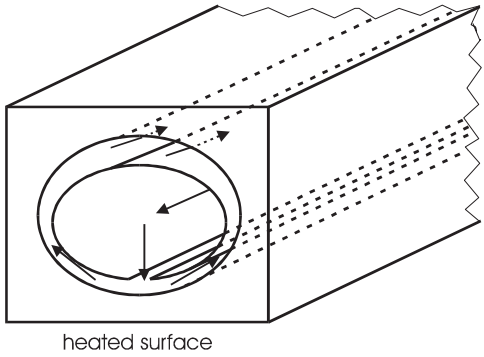


Fig. 15. Confined planar jet as suggested by Inoue et al. [70]. Water is fed through the inner tube, forms a planar jet through the slit in the bottom, and then returns through the outer tube.

For densely packed cells under high concentrations (>150 suns), a thermal resistance of less than $10^{-4}\,K\,m^2/W$ is necessary (see Section 2.3). Only jets and microchannels have reported such low values. Two-phase forced convection could also be a viable alternative. However, this solution would probably require the use of fluids other than water, which violates the requirements of an open loop system and might involve toxic fluids (see Section 1.1).

6. Conclusion

Cell cooling is an important factor when designing concentrating photovoltaic systems. The cooling system should be designed to keep the cell temperature low and uniform, be simple and reliable, keep parasitic power consumption to a minimum and, if possible, enable the use of extracted thermal heat.

With single-cell geometries, research shows that passive cooling is feasible and the most cost-efficient solution for concentration values of at least 1000 suns provided the cells and lenses are kept small.

Linear concentrators can also be cooled passively, but the heat sinks tend to get very intricate and therefore expensive for concentration values above 20 suns. A heat pipe based solution is one way to increase the passive cooling performance. Different ways of active cooling by water or other coolants have also been found to work well and should be considered for concentration levels above 20 suns.

For densely packed cells, active cooling is the only feasible solution. At high concentrations, the high heat flux makes a low contact resistance from cell to cooling system extremely important. The thermal resistance of the cooling system must be kept below $10^{-4}\,K\,m^2/W$ for concentration levels above 150 suns. New solutions such as microchannels or impinging jets may prove to be good solutions. Microchannels are particularly promising because they have the option of being incorporated in the cell manufacturing process. The costs for large scale production of many of these high performance cooling options are yet to be confirmed.

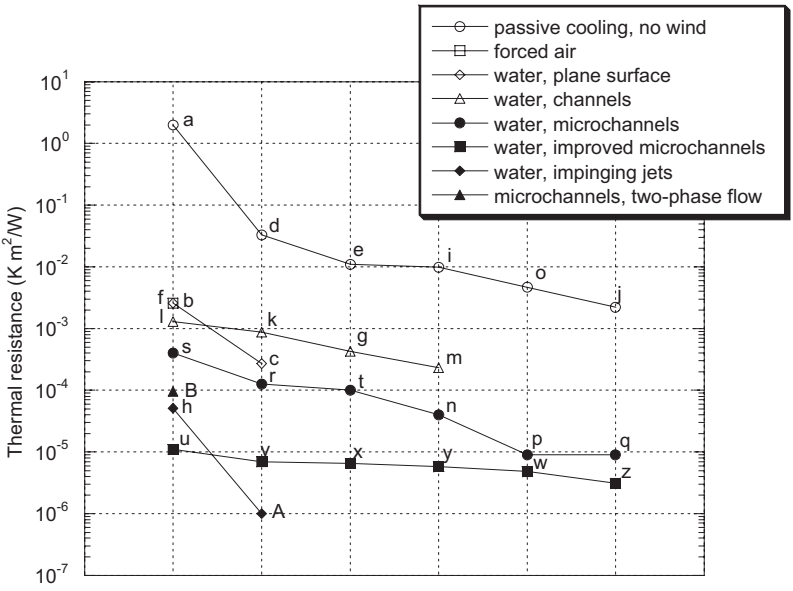


Fig. 16. Comparison of different cooling options. The letters refer to the references listed in Table 2.

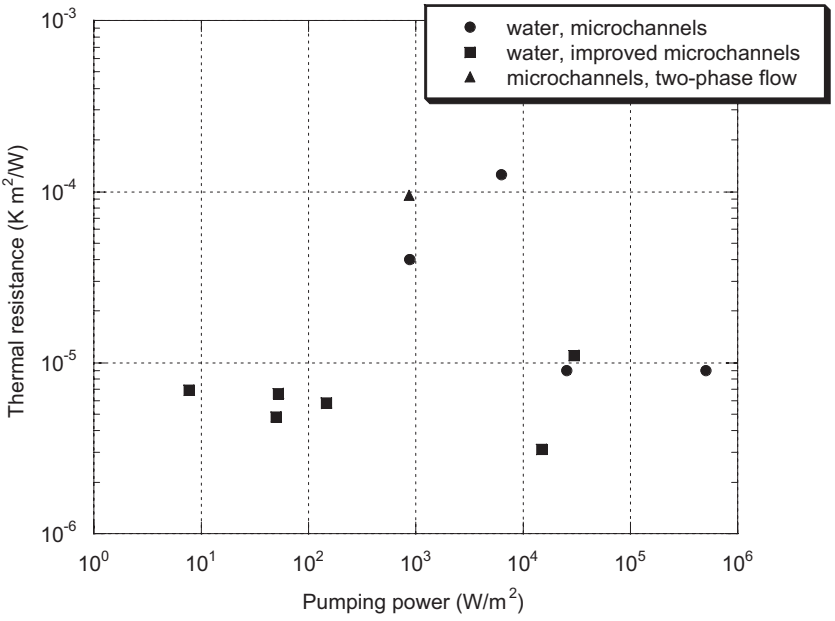


Fig. 17. Comparison of different cooling options and the pumping power they require. The letters refer to the references listed in Table 2.

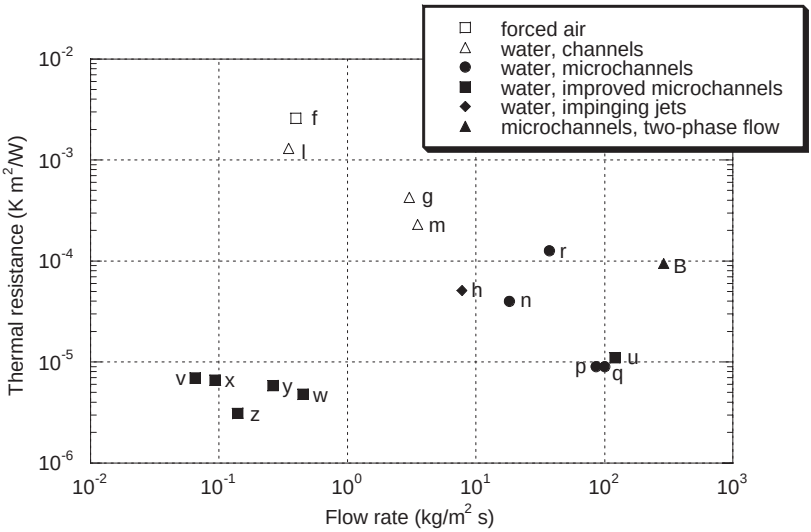


Fig. 18. Comparison of different thermal resistance cooling options and flow rates. The letters refer to the references listed in Table 2.

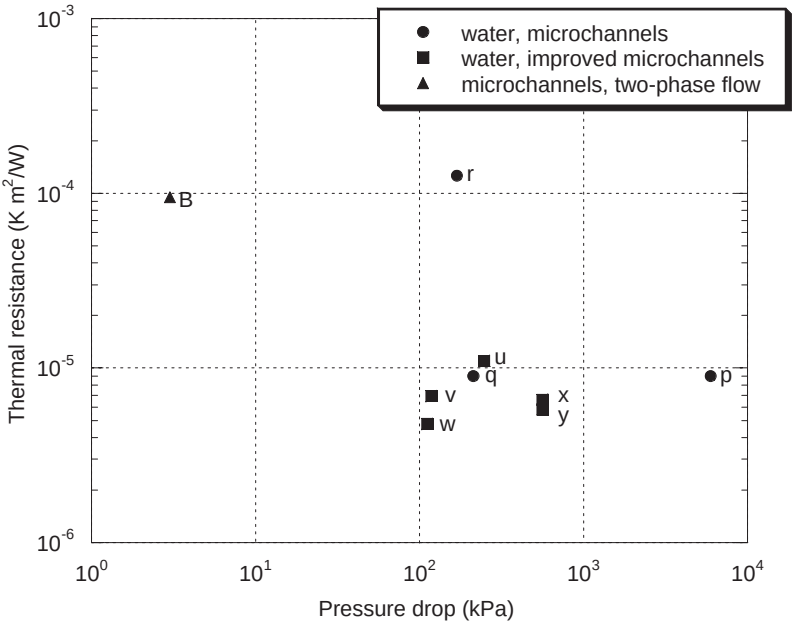


Fig. 19. Thermal resistance versus pressure drop for different cooling options. The letters refer to the references listed in Table 2.

Experimental work is still important for determining the best method of cooling for a given application, but the comparisons in this review may provide a good background.

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